COSC2637 Big Data Processing
Lab 5 - Tutorial/Lab class (week 6)

Task - Python MapReduce program for k-means clustering

Given a set of data points in a two-dimensional space and k locations in the space as the initial cluster centroids, the task is to develop a Python MapReduce program for k-means clustering. The source code from an online source has been provided in course week 6 module on Canvas.

1. Upload following files to AWS EMR cluster mast node. These files include

mapper.py # *mapper*
reducer.py # *reducer*
reader.py # *check whether termination condition is satisfied*
run.sh # *control iterations and termination condition check by reader.py*
dataset.txt # *a set of data points in a 2-dimensional space*
initial_centroids_backup.txt # *contains the initial 3 cluster centroids*

```
[hadoop@ip-192-168-26-208 ~]$ chmod +x run.sh
```

make sure run.sh is executable "chmod +x run.sh"

```
[hadoop@ip-192-168-26-208 ~]$ hadoop fs -copyFromLocal dataset.txt /
```

copy dataset.txt to folder "/" on HDFS by "hadoop fs -copyFromLocal dataset.txt /"

```
[hadoop@ip-192-168-26-208 ~]$ cp initial_centroids_backup.txt centroids.txt
[hadoop@ip-192-168-26-208 ~]$ cp initial_centroids_backup.txt centroids1.txt
```

centroids.txt # *initialized by initial_centroids_backup.txt, and contains results after iterations.*
centroids1.txt # *the centroids after one iteration, used for checking termination condition in run.sh*

```
[hadoop@ip-192-168-26-208 ~]$ cat centroids.txt
```

The content is

```
-10, -40
0, 0
10, 30
```

Now you should have all files ready for execution

```
[hadoop@ip-192-168-25-168 ~]$ ls
centroids.txt   dataset.txt                    mapper.py  reducer.py
centroids1.txt  initial_centroids_backup.txt   reader.py  run.sh
```

Execute run.sh

```
[hadoop@ip-192-168-26-208 ~]$ ./run.sh
```

If running successfully, check the content of centorids.txt

```
[hadoop@ip-192-168-26-208 ~]$ cat centroids.txt
```

You will see something like below

```
-15.000500407, -34.9999749295
1.9999175124, -9.00020832026
9.99997035758, 24.9999516687
```

Read through run.sh (below) to understand the flow the k-means algorithm. The detailed explanation of the code is in report.pdf

```
#!/bin/bash

# Remove the HDFS output directory forcefully and recursively.
hadoop fs -rm -r -f /mapreduce-output*

i=1
while :
do
    hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
    -D mapreduce.job.reduces=3 \
    -D mapred.text.key.partitioner.options=-k1 \
    -files ./centroids.txt,./reducer.py,./mapper.py \
    -mapper ./mapper.py \
    -reducer ./reducer.py \
    -input /dataset.txt \
    -output /mapreduce-output$i \
    -partitioner org.apache.hadoop.mapred.lib.KeyFieldBasedPartitioner

    rm -f centroids1.txt
    hadoop fs -getmerge /mapreduce-output$i/part* centroids1.txt

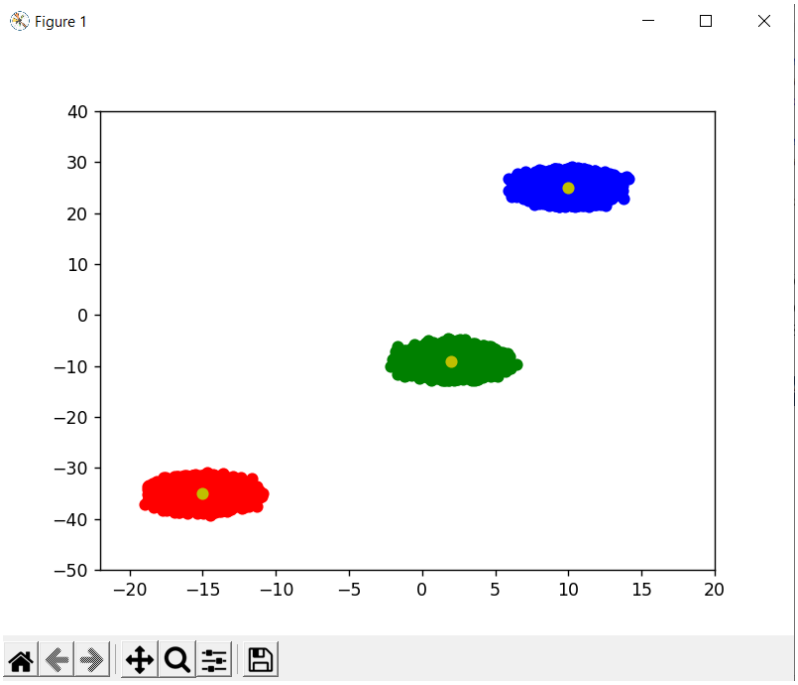
    seeiftrue=`python reader.py`

    if [ $seeiftrue = 1 ]
    then
        rm centroids.txt
        hadoop fs -getmerge /mapreduce-output$i/part* centroids.txt
        break
    else
        rm centroids.txt
        hadoop fs -getmerge /mapreduce-output$i/part* centroids.txt
    fi
    i=$((i+1))
done
```

You can visualize the data points dataset.txt and the cluster centroids results.txt with printer.py (you need make a copy of "results.txt" and name the copy as "centroids.txt"). Put centroids.txt and dataset.txt on local machine (your laptop) in the same folder as printer.py. On my windows, I use "powershell". Run printer.py as:

```
PS H:\Ke\All\teaching\2022\Big Data\week 6\lab\hadoop-kmeans-master\src\MapReduce> python printer.py
```

The output will be:



Exercise: You can create a set of data points in a 2-dimensional space and perform k-means clustering. You may find the location of initial clustering centroids are essential. Why?

Don't forget to terminate your cluster at end of the lab class!

Also, if trying out any these exercises outside of the lab class, make sure you terminate your cluster and DO NOT leave clusters running idle for days.

```
./terminate_cluster.sh
```